

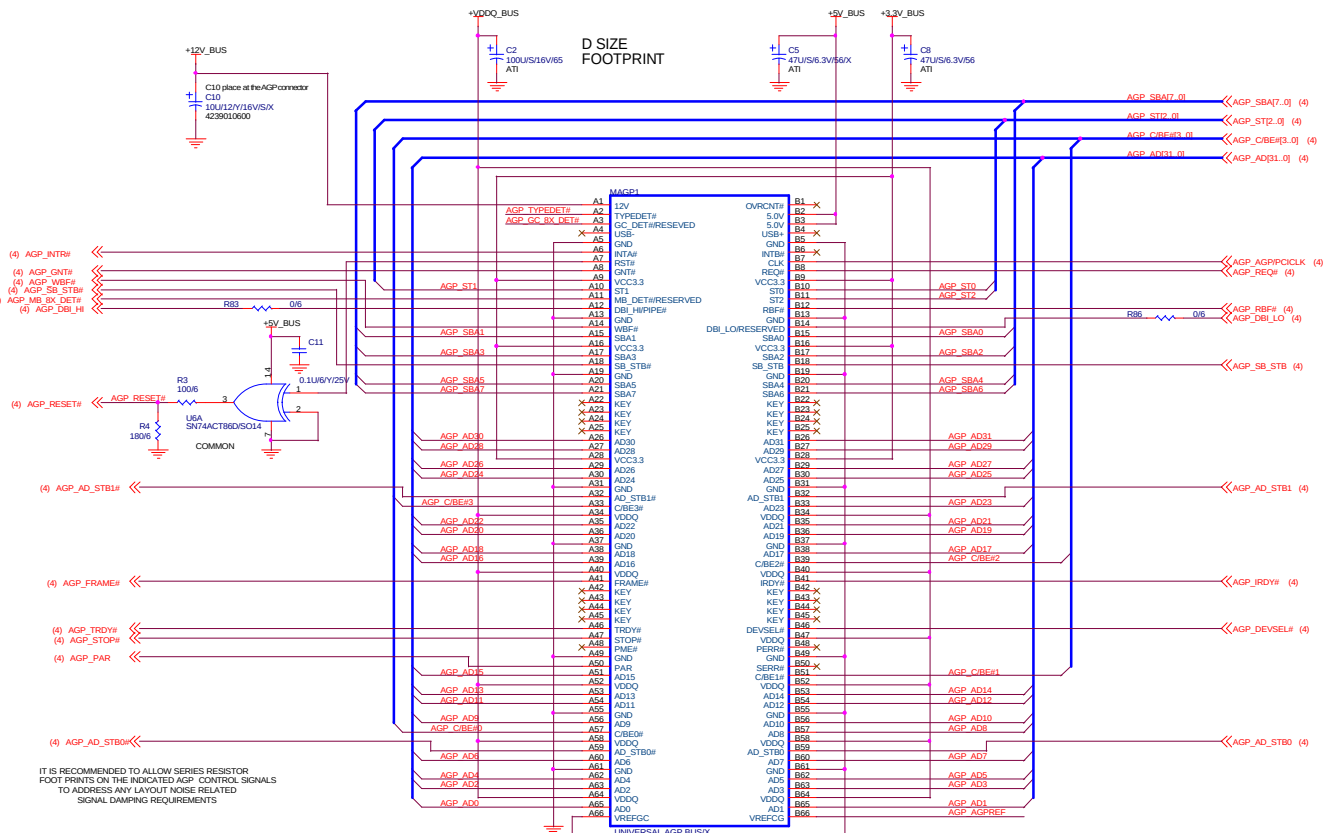
GIGABYTE GV-R9200NF-NF3 Schematics

TITLE SHEET

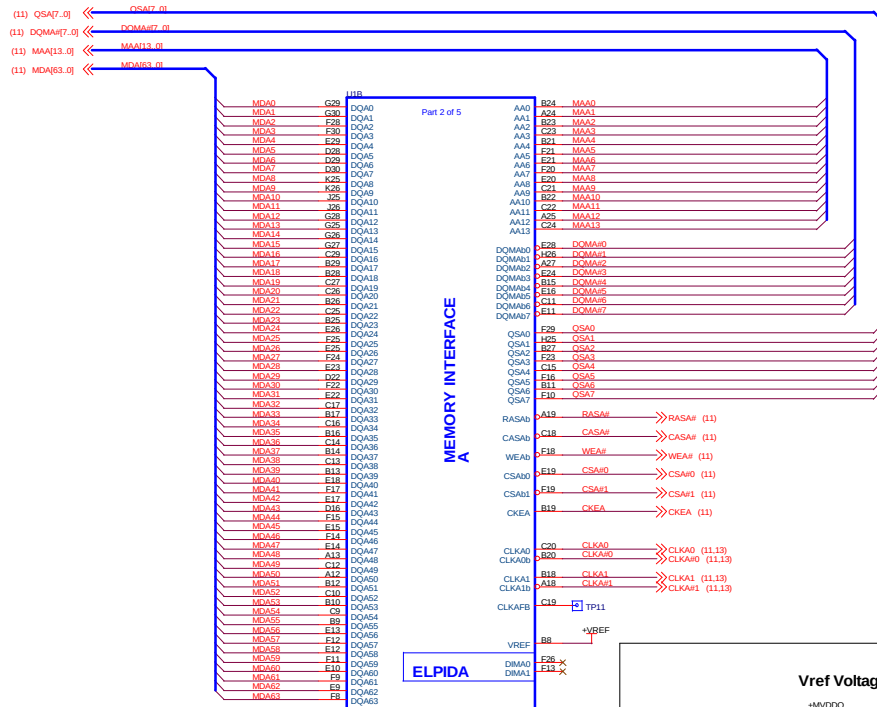
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The following grounds should be routed back to their respective regulators and then tied directly to the ground plane with one via: GND_PVSS, GND_MPVSS, GND_TPVS, and GND_A2VSSN. The other ground pins (GND_AVSSN, GND_A2VSSQ, GND_RSET, GND_R2SET) should be tied to the ground plane directly through one via as close to the pins as possible without connecting to anything else. If space is an issue it is possible to use one via for two adjacent pins.

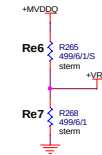
D SIZE FOOTPRINT



MEMORY CHANNEL A

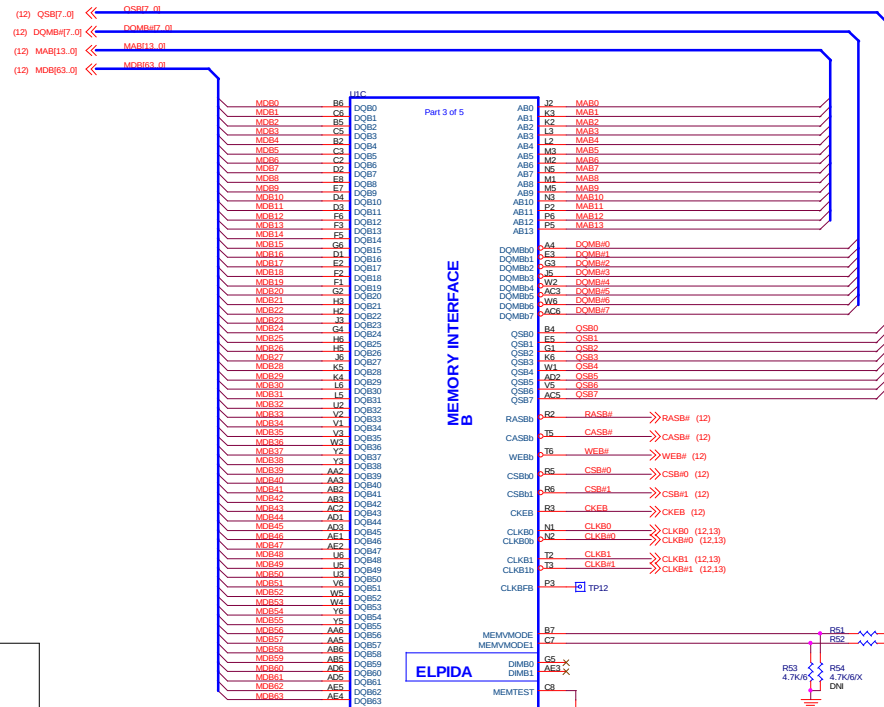


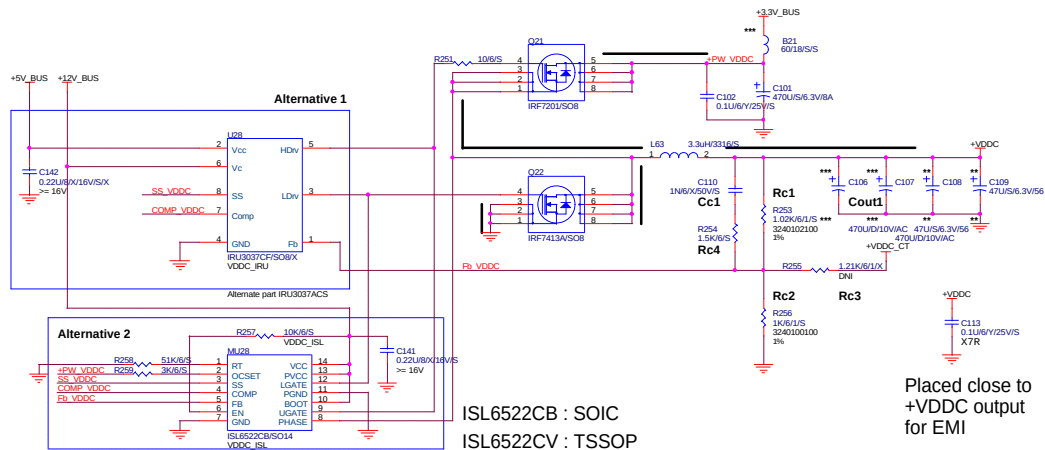
Vref Voltage



Use localized Vref on the memory page

MEMORY CHANNEL B

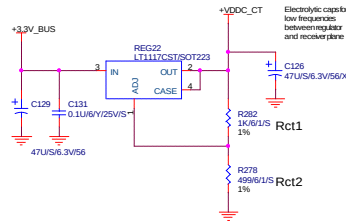




Regulator for VDDC_CT (Core Transform) and AVDD/A2VDDQ/AVDDDD/A2VDDDDI/TXVDDR, LVDDR, MPVDD

Vin = 3.3V AGP
Vout = 1.8V
Iout = 350mA + 100mA + 50mA = 500mA MAX
Iout = 600mA MAX (with PVDD/TPVDD)

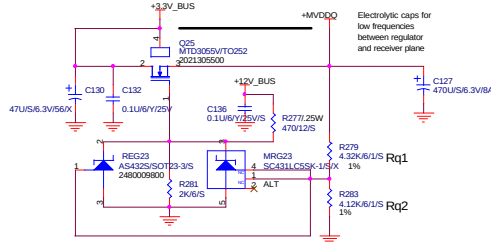
	Rct1		Rct2	
1.8V	1K	3240100100	603	422R 3240422000 603
1.9V				499R 3240499000 603



Regulator for MVDDQ (MEM IO) and VDDR1

Vin = 3.3V AGP
Vout = 2.5V (TSOP)
Iout = 1200mA MAX
Iout = 1000mA Est. MAX

Q25 Pin2/4 should be soldered to board for heat dissipation and a GND fill area.

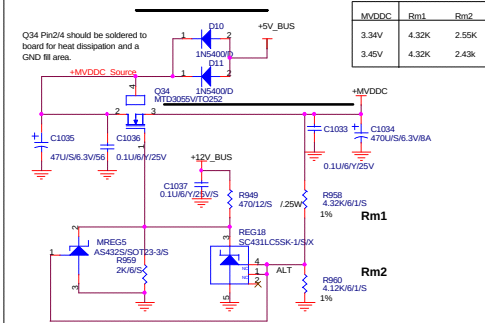


Regulator for MVDDC

Vin = 5V
Vout = 3.4V 2A MAX

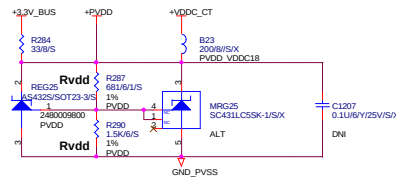
These dummy resistors are placed under the diodes to avoid PCB heat damage due to hot diodes.

R1261 012	R1262 012	R1263 012	R1264 012
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Regulator for PVDD (Core PLLs) and optional TPVDD (TMDS PLLs)

Vin = 3.3V AGP
Vout = +1.8V
Iout = 25mA MAX (PVDD only)
Iout = 30mA MAX (PVDD + TPVDD)



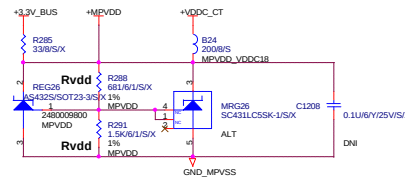
The value of resistor were chosen to reduce failure rate caused by possible defective regulators, i.e., 33R are used instead of 47R or 51R for more start up current. (3.465V - 1.8V) / 33R = 50.5mA

805 package resistor are required for sufficient power rating (0.1W rating). (3.465V - 1.8V) * 50.5mA = 0.085W; therefore, smaller resistor value would require 1206 package

Regulator for MPVDD (Memory PLLs)

Vin = 3.3V AGP
Vout = +1.8V
Iout = 10mA MAX

(Optional)



Regulator For TPVDD (TMDS PLLs)

Vin = +3.3V AGP
Vout = 1.8V
Iout = 15mA MAX

TPVDD might not be needed if PVDD can provide stable 1.8V

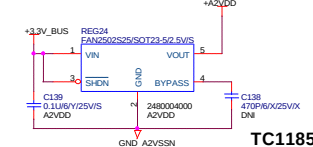
(Optional)



Regulator For A2VDD (2nd DACs)

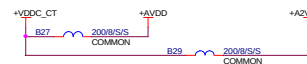
Vin = +3.3V AGP
Vout = 2.5V
Iout = 150mA MAX

A2VDD might not be needed if VDD can provide stable 2.5V



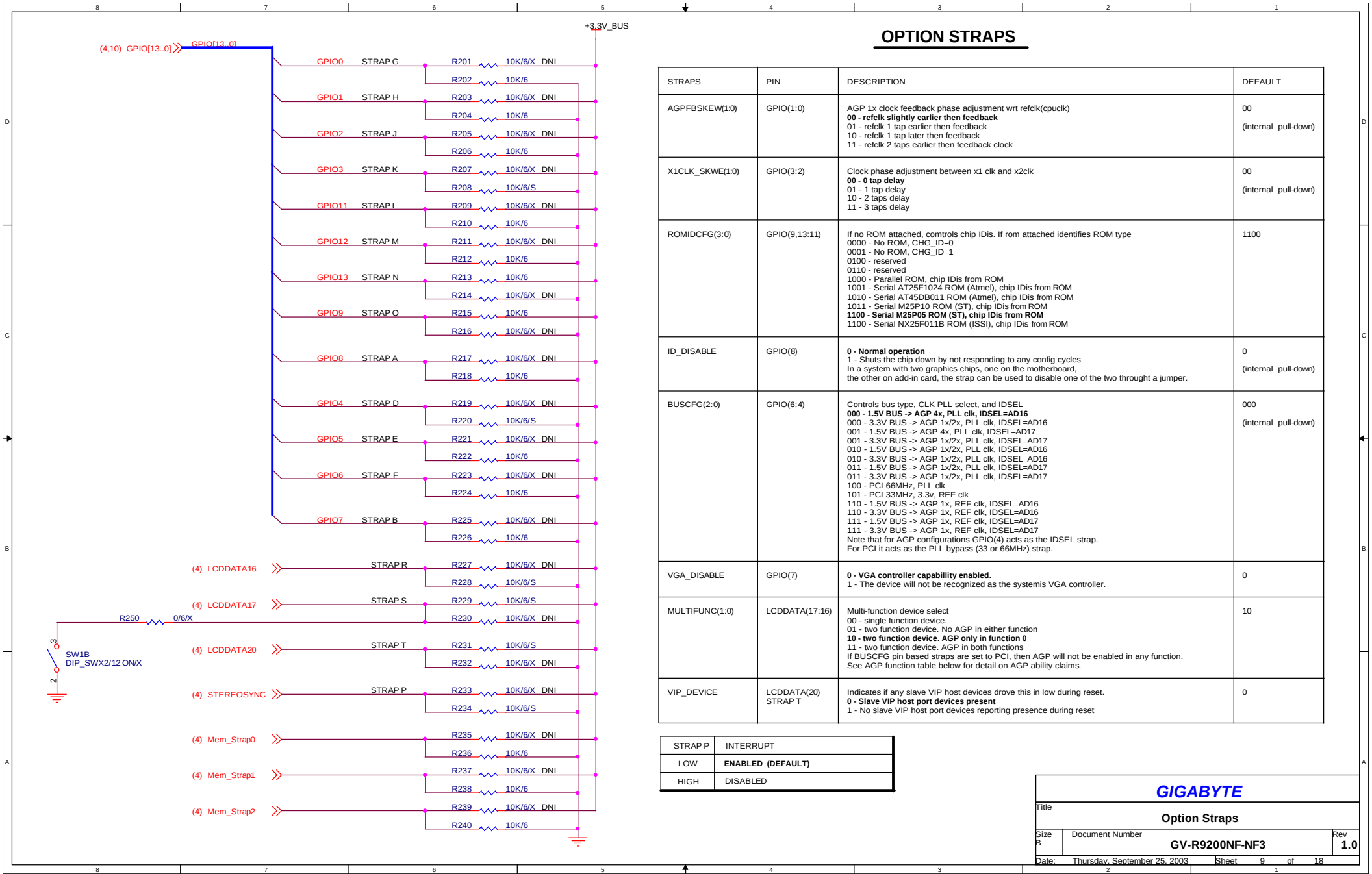
A2VDD and A2VSSN routed with at least 15 mil trace and not longer than 1.5 inch. A2VSSN with signal via to GND at the regulator

AVDD/A2VDDQ (1st DAC & 2nd DAC Band Gap)



Some Part Ref's updated to 988 brd

GIGABYTE			
Power_REG (VDDQ/etc)			
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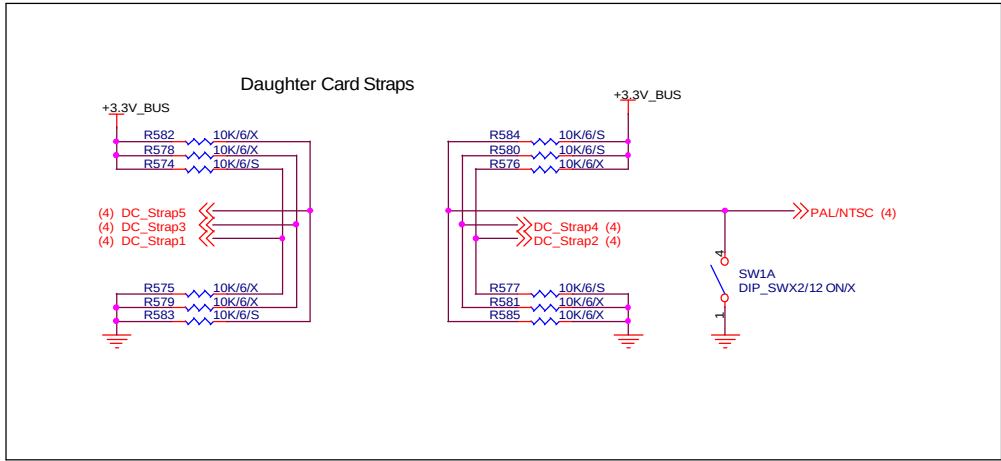
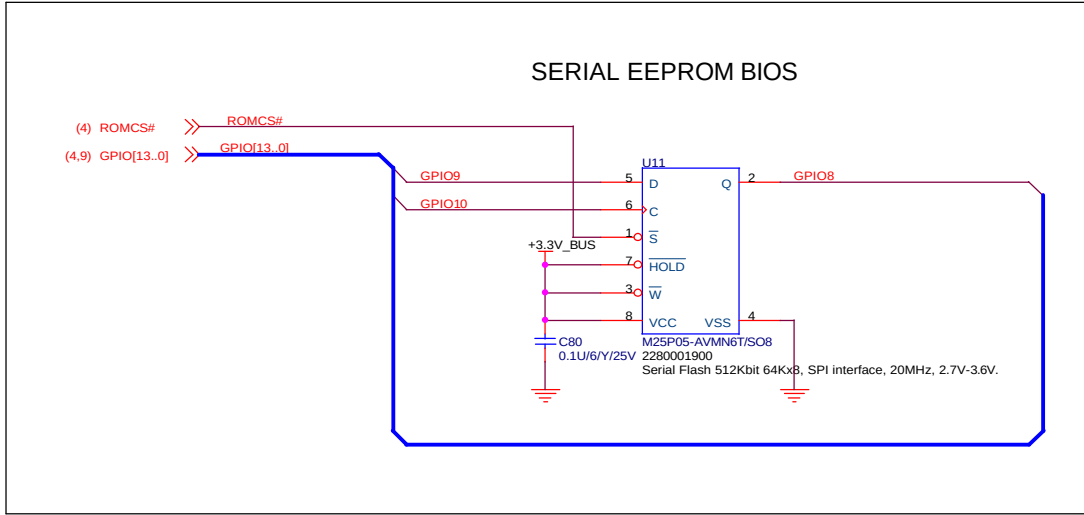


OPTION STRAPS

STRAPS	PIN	DESCRIPTION	DEFAULT
AGPFBSKEW(1:0)	GPIO(1:0)	AGP 1x clock feedback phase adjustment wrt refclk(cpuckl) 00 - refclk slightly earlier then feedback 01 - refclk 1 tap earlier then feedback 10 - refclk 1 tap later then feedback 11 - refclk 2 taps earlier then feedback clock	00 (internal pull-down)
X1CLK_SKWE(1:0)	GPIO(3:2)	Clock phase adjustment between x1 clk and x2clk 00 - 0 tap delay 01 - 1 tap delay 10 - 2 taps delay 11 - 3 taps delay	00 (internal pull-down)
ROMIDCFG(3:0)	GPIO(9,13:11)	If no ROM attached, controls chip IDIs. If rom attached identifies ROM type 0000 - No ROM, CHG_ID=0 0001 - No ROM, CHG_ID=1 0100 - reserved 0110 - reserved 1000 - Parallel ROM, chip IDIs from ROM 1001 - Serial AT25F1024 ROM (Atmel), chip IDIs from ROM 1010 - Serial AT45DB011 ROM (Atmel), chip IDIs from ROM 1011 - Serial M25P10 ROM (ST), chip IDIs from ROM 1100 - Serial M25P05 ROM (ST), chip IDIs from ROM 1100 - Serial NX25F011B ROM (ISSI), chip IDIs from ROM	1100
ID_DISABLE	GPIO(8)	0 - Normal operation 1 - Shuts the chip down by not responding to any config cycles In a system with two graphics chips, one on the motherboard, the other on add-in card, the strap can be used to disable one of the two through a jumper.	0 (internal pull-down)
BUSCFG(2:0)	GPIO(6:4)	Controls bus type, CLK PLL select, and IDSEL 000 - 1.5V BUS -> AGP 4x, PLL clk, IDSEL=AD16 000 - 3.3V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD16 001 - 1.5V BUS -> AGP 4x, PLL clk, IDSEL=AD17 001 - 3.3V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD17 010 - 1.5V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD16 010 - 3.3V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD16 011 - 1.5V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD17 011 - 3.3V BUS -> AGP 1x/2x, PLL clk, IDSEL=AD17 100 - PCI 66MHz, PLL clk 101 - PCI 33MHz, 3.3v, REF clk 110 - 1.5V BUS -> AGP 1x, REF clk, IDSEL=AD16 110 - 3.3V BUS -> AGP 1x, REF clk, IDSEL=AD16 111 - 1.5V BUS -> AGP 1x, REF clk, IDSEL=AD17 111 - 3.3V BUS -> AGP 1x, REF clk, IDSEL=AD17 Note that for AGP configurations GPIO(4) acts as the IDSEL strap. For PCI it acts as the PLL bypass (33 or 66MHz) strap.	000 (internal pull-down)
VGA_DISABLE	GPIO(7)	0 - VGA controller capability enabled. 1 - The device will not be recognized as the system's VGA controller.	0
MULTIFUNC(1:0)	LCDDATA(17:16)	Multi-function device select 00 - single function device. 01 - two function device. No AGP in either function 10 - two function device. AGP only in function 0 11 - two function device. AGP in both functions If BUSCFG pin based straps are set to PCI, then AGP will not be enabled in any function. See AGP function table below for detail on AGP ability claims.	10
VIP_DEVICE	LCDDATA(20) STRAP T	Indicates if any slave VIP host devices drove this in low during reset. 0 - Slave VIP host port devices present 1 - No slave VIP host port devices reporting presence during reset	0

STRAP P	INTERRUPT
LOW	ENABLED (DEFAULT)
HIGH	DISABLED

GIGABYTE			
Title Option Straps			
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GIGABYTE			
Title			
Serial BIOS & HW Straps			
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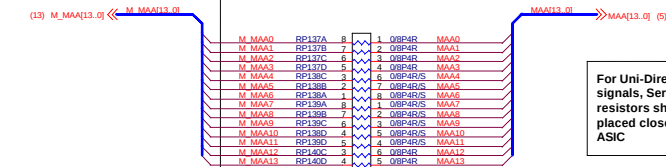
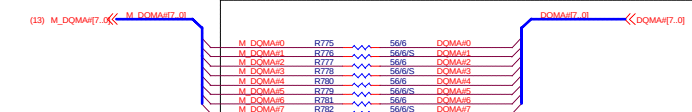
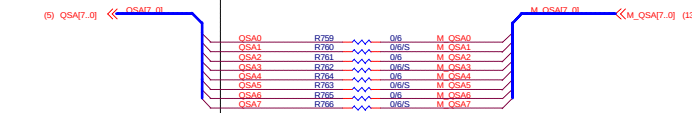
TERMINATION FOR MEMORY CHANNEL A



SERIES Resistors

For Bi-Directional signals, Series resistors should be placed close to the memory

Proper Termination of QSA?

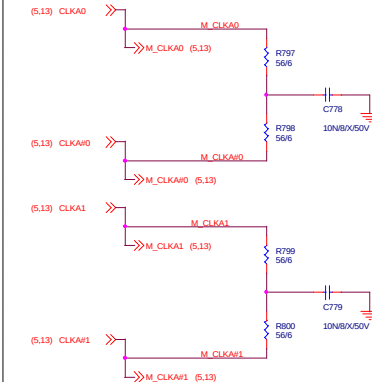


For Uni-Directional signals, Series resistors should be placed close to the ASIC

CLOCK terminations

Change from 1:1 spacing to at least a 2.5:1 spacing between the pair

These resistors and caps must be placed to minimize any stubs. These must also be placed after the memory



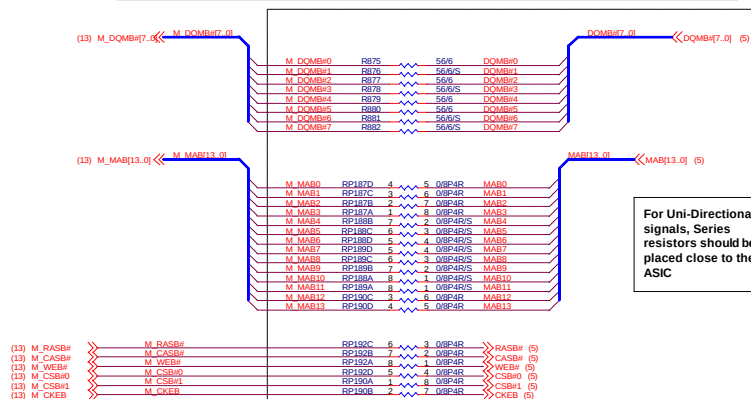
GIGABYTE

Serial Termination CH A		
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TERMINATION FOR MEMORY CHANNEL B



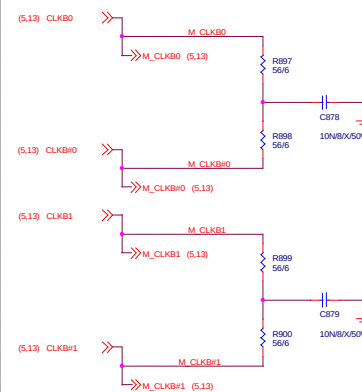
Proper Termination of QSB?



CLOCK terminations

Change from 1:1 spacing to at least a
2.5:1 spacing between the pair

These resistors and caps must be placed to minimize any stubs. These
must also be placed after the memory



GIGABYTE

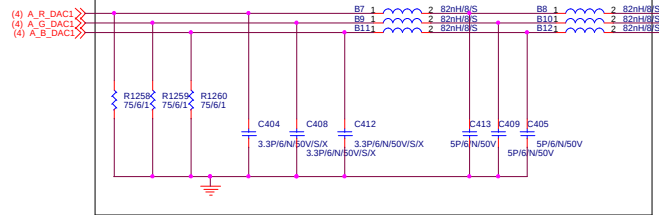
Serial Termination CH B

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PRIMARY CRT

Place close to ASIC

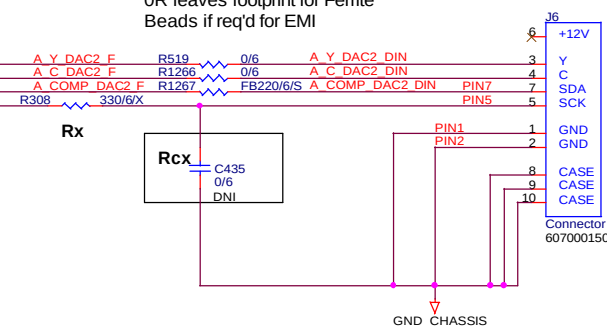
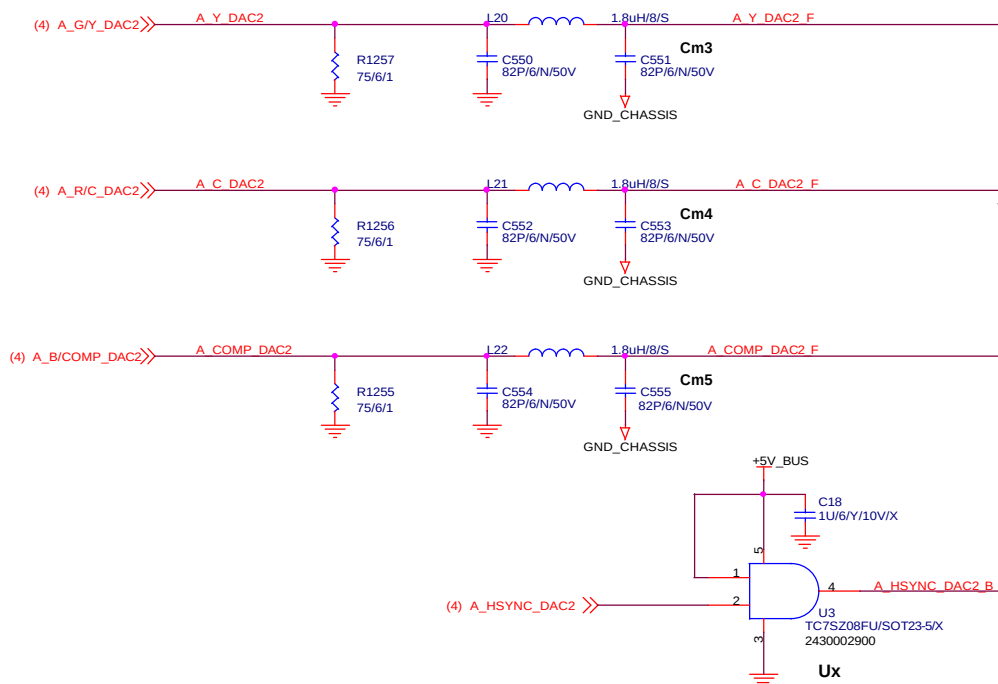


Place Resistors close to ASIC.

Add alternate part for 7 pin Svideo
6071001500

TV Out (SVHS)

Place near connector
0R leaves footprint for Ferrite
Beads if req'd for EMI

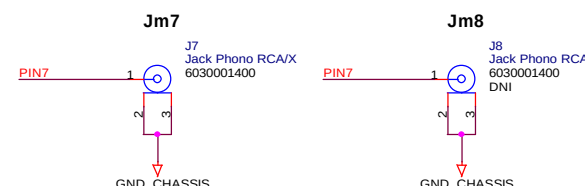


HSync to Connector	Rx = 330R, Rcx = 470pF, Ux INSTALL
HSync NOT Connected	Rx = DNI, Rcx = 0R, Ux = DNI

TV Out (Comp)

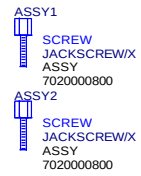
Jm6 and Jm7 use
the same footprint

DO NOT POPULATE
for footprint only



GIGABYTE			
Title Video MUX & VIVO			
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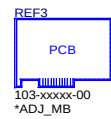
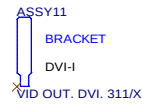
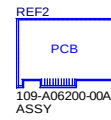
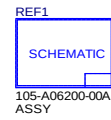
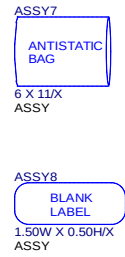
CRT SCREWS



DVI SCREWS



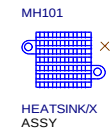
MISC. BOARD PARTS



Heatsink

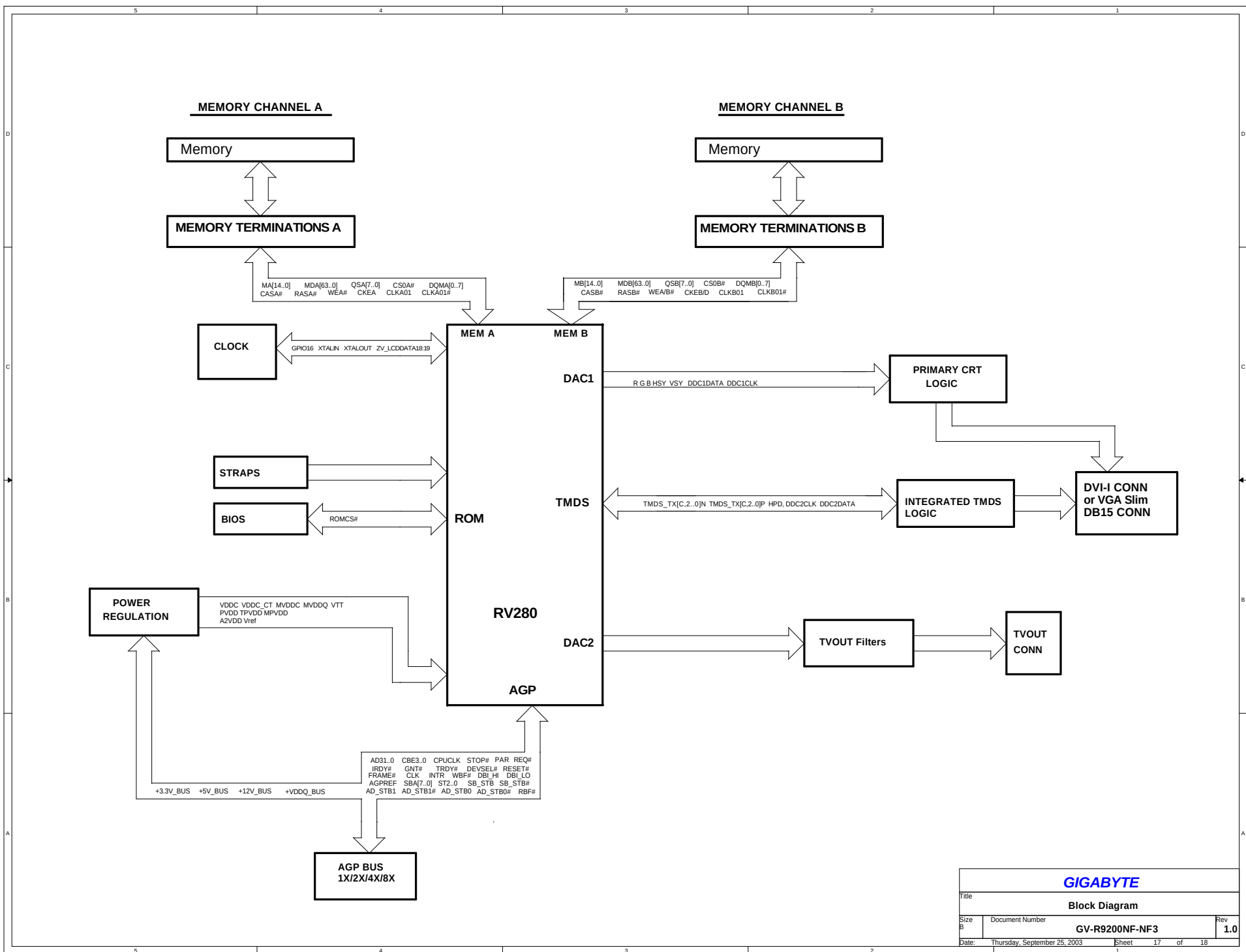
**PASSIVE
HEATSINK**

H103
HEATSINK/X
ALT
7120000700
Heatsink cross-hatched finned
45X45X10mm black amodize w/o
adhesive



GIGABYTE

Title		
Heatsink & Mechanicals		
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5		4		3		2		1	
		Title RV280 AGP8x 32/64/128MB 4/8x16 DDR				Schematic No. GV-R9200NF-NF3		Date: Thursday, September 25, 2003	
REVISION HISTORY								Rev 1.0	
Sch Rev	Date	REVISION DESCRIPTION							
0	09/21/02	PRELIMINARY based on RV250 REFERENCE REF109 Schematic and RV250 BGA 958 Schematic							
0.1	09/23/02	PRELIMINARY based on the RV280 REFERENCE REF 105-REF112-00A							
	09/24/02	Copied from 09/24/02 version 0 of 105-ref112-00a RV280 Reference Design - pg1 Add AGP8X cold boot pull-ups - pg9,10 Move DQM Memory Termination Resistor close to ASIC							
	09/25/02	Ready for design review. open issues listed below.							
	09/25/02	John Digweed updated the ASIC to 215R78ANA11H							
	9/27/02	MVDD circuit changed to linear, based on 988 brd.							
	10/01/02	QSMA4/5 corrected. - (pg 1) Rearrange XOR gates, eliminate single AND gate - (pg 1) Rename AGP_AD_STB1# to AGP_AGPREF (cold boot fix) - (pg 6) Remove A2VDD option from +MVDDQ - (pg 7) Add resistor to disable microswitch							
	10/03/02	Branch from filename 105-A06200-00A netlisted_1002, renamed part refs to match 105-988000-00A							
	10/04/02	- (pg 1) Add TXVSSR ground symbol connected to digital GND - (pg 1) Change VDDQ_BUS cap to 47uF tant - (pg 1) Remove VREFGC fixed resistor/capacitor circuit, change R91 to 1K - (pg 1) Change value of C76 capacitor (AGP_AGPREF) - (pg 4) Replace 1uF cap on TXVDDR to 100pF and 100nF X7R caps - (pg 4) Replace 10uF cap on A2VDD to 22uF electrolytic or 10uF tantalum cap - (pg 4) Add 10nF capacitor array on +VDDQ_BUS - (pg 5) Update VDDC Alt1 regulator circuits - (pg 14) Change 0R for chassis and digital ground to 805 resistors - (pg 1) Removed AGP 8x Cool Boot Fix - (pg 1) Removed Q8, R87 for AGP_REFCG							
	10/07/02	- (pg 1) C2, C5, C8 changed to 100 uF							

GIGABYTE

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